WORKED SOLUTION

STEP III 2013, Question 8

Pure • Polar Equations of Conics

STEP III 2013, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q8

The key idea

The relation $r = ks$ with $k < 1$ says that E is an ellipse with focus O and directrix D, and in polar coordinates centred on O it becomes a very simple equation for r in terms of θ. A chord through O is just two radii pointing in opposite directions, so $1/l$ comes out as a simple expression in $\cos^2\theta$. The n equally spaced lines then turn the sum into a sum of $\cos 2\theta$ terms, which is exactly what the opening sum disposes of.

Solution

The opening sum

The terms form a geometric progression with first term $e^{2i\alpha}$ and common ratio $w = e^{2\pi i/n}$. Since $n \geq 2$, the angle $2\pi/n$ lies in $(0, \pi]$, so $w \neq 1$ and we may use the usual formula:

$$\sum_{r=0}^{n-1} e^{2i(\alpha + r\pi/n)} = e^{2i\alpha}\,\frac{1 - w^n}{1 - w} = e^{2i\alpha}\,\frac{1 - e^{2\pi i}}{1 - w} = 0.$$

Taking real parts, $\displaystyle\sum_{r=0}^{n-1} \cos(2\alpha + 2r\pi/n) = 0$; this is the form we shall use.

The distance s

Measure θ from the perpendicular from O to D, as in the diagram. The foot of that perpendicular is at distance d from O, and P is at distance $r\cos\theta$ from O in the same direction, so for P on the same side of D as O,

$$s = d - r\cos\theta.$$

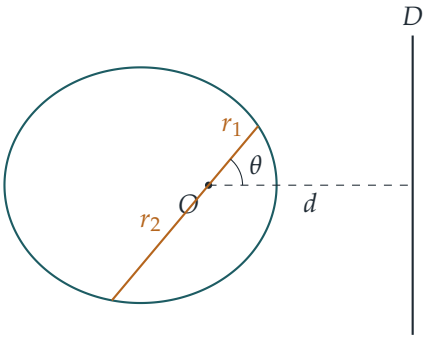
(This holds for every θ: when $\cos\theta < 0$, P is further from D than O is, and the formula correctly gives $s > d$.)

The polar equation of E

Every point of E is on the same side of D as O. Indeed, if P were on the other side we would have $s = r\cos\theta - d$, and $r = ks$ would give $r(1 - k\cos\theta) = -kd < 0$, which is impossible because $1 - k\cos\theta > 0$ for $0 < k < 1$. So we can use $s = d - r\cos\theta$, and $r = ks$ becomes

$$r = k(d - r\cos\theta) \quad \Longrightarrow \quad r = \frac{kd}{1 + k\cos\theta}.$$

The denominator is positive for every θ since $k < 1$, so each ray from O meets E exactly once.



The length of a chord through O

A line through O making angle θ with the perpendicular meets E at the points in directions θ and $\theta + \pi$, which lie on opposite sides of O. Since $\cos(\theta + \pi) = -\cos\theta$, the chord length is

$$l = \frac{kd}{1 + k\cos\theta} + \frac{kd}{1 - k\cos\theta} = \frac{2kd}{1 - k^2\cos^2\theta}, \qquad \text{so} \qquad \frac{1}{l} = \frac{1 - k^2\cos^2\theta}{2kd}.$$

Note that this depends only on the line, not on which of its two directions we call θ.

? **Ask yourself** How do we sum $\cos^2\theta$ over the n lines?

Use the double-angle formula $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$: the $\cos 2\theta$ terms are exactly the real parts of the opening sum.

The sum over the n lines

Since adjacent lines are at angle π/n, the n lines make angles $\theta_j = \alpha + j\pi/n$ $(j = 1, \ldots, n)$ for some fixed α. These n values of $2\theta_j$ are $2\alpha + 2j\pi/n$; reducing $j = n$ to $j = 0$ (which changes $2\theta_j$ by 2π) they are the angles of the opening sum. Hence

$$\sum_{j=1}^{n} \cos^2\theta_j = \frac{n}{2} + \frac{1}{2}\sum_{j=1}^{n} \cos 2\theta_j = \frac{n}{2} + 0 = \frac{n}{2},$$

and so

$$\sum_{j=1}^{n} \frac{1}{l_j} = \frac{1}{2kd}\left(n - k^2 \cdot \frac{n}{2}\right) = \frac{(2-k^2)n}{4kd},$$

which is the required result. The condition $n \geq 2$ was used exactly once, to ensure $w \neq 1$ in the opening sum.

Checks

- Take $n = 2$, $k = \frac{1}{2}$, $d = 1$ with lines along and perpendicular to OD. Along OD the chord is the major axis, of length $\frac{1/2}{3/2} + \frac{1/2}{1/2} = \frac{4}{3}$; perpendicular to it the chord (the latus rectum) has length $2kd = 1$. So $\frac{3}{4} + 1 = \frac{7}{4}$, and the formula gives $\frac{(7/4)\cdot 2}{2} = \frac{7}{4}$.
- For $n = 1$ the result fails, as it should: a single line's $1/l$ depends on its direction. The case $n = 1$ is exactly where $w = 1$.
- A computer check, finding each chord directly from $r = ks$ for random d, k, n and α, agreed with the formula; for example $d = 1$, $k = \frac{1}{2}$, $n = 3$ gives $2.625 = \frac{21}{8}$ for any α.

Common slips

- Writing $s = d + r\cos\theta$: in the diagram θ is measured towards D, so moving out along OP brings you closer to D.
- Taking the second half of the chord as $kd/(1 + k\cos\theta)$ again, forgetting that the opposite direction has angle $\theta + \pi$.
- Summing the angles $\alpha + j\pi/n$ directly instead of $2\theta_j$: it is the doubled angles that go once round the circle evenly, which is why the opening sum has the factor 2.

Beyond STEP

The focus–directrix definition gives every conic the polar equation $r = \ell/(1 + e\cos\theta)$, with $e < 1$, $e = 1$ and $e > 1$ for ellipse, parabola and hyperbola; this is the form in which Kepler's orbits appear when one solves Newton's equations for planetary motion. The fact that $1/l_1 + 1/l_2$ is the same for any two perpendicular focal chords is a classical property, and the result here generalises it. Kleppner and Kolenkow, *An Introduction to Mechanics*, derives the orbit equation this way.

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